

Exploring the Potential of Machine Learning and Deep Learning in ECG Image Analysis for Cardiovascular Disease Diagnosis

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Abstract—This research proposes a novel machine learning approach to detect cardiovascular diseases (CVDs) in electrocardiogram (ECG) images. The goal is to improve diagnostic accuracy and efficiency within the healthcare system. CVDs remain the leading cause of death globally, emphasizing the importance of early and accurate diagnosis for effective treatment. ECG signals provide valuable information for assessing cardiovascular health. The methodology utilizes advanced machine learning algorithms for classification. First, it pre-processes ECG signals to extract relevant features. The results demonstrate the machine learning model's potential as a significant diagnostic tool by showcasing its accuracy in detecting various cardiovascular illnesses from ECG images. Specifically, this research leveraged the power of machine learning to predict four major heart abnormalities: arrhythmias, myocardial infarction (heart attack), heart failure, and hypertrophic cardiomyopathy. This investigation explores how different machine learning methods perform when analyzing ECG images to identify cardiovascular diseases. The primary objective is to identify the algorithm that delivers the highest accuracy in classifying ECG images for CVD detection. Decision trees are best suited for working with data organized in tables, where each entry has specific numerical values or categories. Random forests, like decision trees, are primarily designed to handle data in a tabular format with features like numbers or categories. ECG image analysis necessitates techniques adept at recognizing intricate patterns within the pixels of an image. The ultimate goal is to identify the most effective machine learning approach for accurate CVD detection using ECG images, potentially leading to improved healthcare system efficiency and patient outcomes. In the proposed system the algorithm with the highest accuracy for CVD diagnosis using ECG image analysis is the Decision Tree, achieving 59.36% accuracy.

Keywords—Cardiovascular diseases(CVD), Support Vector Machine(SVM), Decision Tree(DT), Random Forest(RF), Deep Learning , Machine Learning.

I. INTRODUCTION

Cardiovascular diseases (CVDs) represent the leading worldwide factor contributing to death, causing around 17.9 million fatalities each year. This category encompasses a wide range of heart and blood vessel problems, including conditions that affect the arteries supplying blood to the heart (coronary heart disease), blood flow to the brain (cerebrovascular disease), and the valves within the heart (rheumatic heart disease). A staggering majority, over four out of five, of cardiovascular disease deaths are caused by heart attacks and strokes. Shockingly, a third of these deaths strike people younger than 70, cutting their lives tragically short. Efficient detection of cardiovascular disease at an earlier stage could save numerous lives. Various techniques are employed in the healthcare system for detecting heart diseases, including electrocardiogram (ECG), echocardiography (echo), cardiac magnetic resonance imaging, computed tomography, blood tests. It is employed to identify cardiovascular diseases related to the heart [1]. Although experienced doctors can analyze ECG patterns to detect heart disease, this traditional method can be prone to errors and is quite slow. The application of machine learning and deep learning techniques has become instrumental in the automatic prediction of heart diseases. Identifying key characteristics from ECG data is crucial for using computers to automatically analyze and predict heart disease. The goal of feature extraction is to identify and represent the most informative aspects of the data that are relevant to the prediction of heart diseases. ECG is a diagnostic

tool[2] that records the electrical activity of the heart over time, and abnormalities in the signal can indicate various cardiovascular conditions. CNNs are well-suited for image-based tasks, and they can be applied to 2D representations of ECG signals. In this investigation, The deep CNN demonstrates a remarkable ability to distinguish between normal and abnormal ECG patterns, achieving an accuracy of 94.95%, sensitivity of 93.72%, and specificity of 95.18% for Net 1 (two seconds). For Net 2 (5 seconds), the accuracy reaches 95.11%, with a sensitivity of 91.13% and specificity of 95.88%. Convolutional Neural Networks (CNNs)[3] consist of two essential components. In the first part, the system automatically finds important patterns within the information provided. Then, a separate section analyzes these patterns to categorize the data. In the analysis of the dataset presented in, three distinct supervised machine learning algorithms, namely Naive Bayes, K-NN, and the Decision List algorithm, were employed. The Tanagra tool was utilized for data classification, and the evaluation was conducted through 10-fold cross-validation, with the results being systematically compared. This tool facilitates[4] the analysis of both real and synthetic data, offering a robust system encompassing clustering, supervised learning, meta-supervised learning, feature selection, data visualization, supervised learning assessment, statistics, and feature selection and construction algorithms. The Decision Tree, recognized for its simplicity and ease of implementation, is a popular classifier that requires no domain knowledge or parameter setting and can effectively handle high-dimensional data. Naïve Bayes, a statistical classifier assuming no dependency between attributes, seeks to maximize the posterior probability for class determination. Its advantage lies in its applicability without the need for Bayesian methods, and it has demonstrated [5] effectiveness in various complex real-world scenarios. The k-nearest neighbor's algorithm (k-NN), a method for classifying objects based on proximity to training data in the feature space, operates as an instance-based learning approach. The experimental setup involved a training dataset comprising 3000 instances with 14 different attributes. This dataset was split into 70% for training and 30% for testing.

Additional high-performing machine learning classifiers encompass Random Forest, Logistic Regression, SVM, and K-Nearest Neighbors, demonstrating comparable accuracy and precision scores, albeit with slightly reduced recall and F1 scores [6]. Conversely, classifiers like XGBoost and AdaBoost, while achieving moderate accuracy and precision scores, exhibit comparatively lower recall and F1 scores. Research in these domains offers a promising avenue for predicting CVD early, aiding clinicians in decision-making, and reducing fatalities. A pre-trained AlexNet for automatic ECG arrhythmia diagnostics, utilizing it as a feature extractor in tandem with a backpropagation neural network. This approach successfully identifies three specific heart rhythm abnormalities from a large database with an impressive 98.51% accuracy rate. It utilizes a similar method to prior research examining heart disease using short segments of ECG data.

The deep CNN exhibits an accuracy of 94.95% for two seconds and 95.11% for five seconds, aiding clinicians in precise and reliable CAD decision-making based on ECG

signals. The given current data mining techniques for knowledge discovery in healthcare systems, focusing on Heart Disease Prediction. Comparative experiments highlight Decision Tree's superior performance, with Bayesian Classification exhibiting similar accuracy. The experiment utilized a training dataset comprising 3000 instances with 14 attributes, divided into 70% for training and 30% for testing.

II. LITERATURE REVIEW

Neha et. al. [8], proposed three methods, emphasizing the superior performance of machine learning algorithms, particularly KNeighbors classifier, in analysis with a dataset not necessarily large. Efficient data preprocessing reduced computational time, and normalization proved crucial to prevent overfitting. Statistical analysis, Gaussian distribution, and Isolation Forest for outlier detection were highlighted. It explores advancements in deep learning algorithms (DLA) for medical image analysis, covering the history of neural networks, breakthroughs in recent deep learning (DL) algorithms, and various DL techniques like auto-encoders, recurrent, CNN, and restricted Boltzmann machines. Artificial intelligence has been developed [9] to automatically analyze electrocardiogram (ECG) recordings and classify different heartbeats. This method has been shown to be highly accurate, particularly in identifying abnormal heart rhythms. This paves the way for real-time heart monitoring systems that could potentially warn of heart problems. Avanzato et. al [10] presented a simple and memory-efficient network architecture suitable for mobile applications, offering improved performance on the ImageNet dataset and outperforming real-time detectors for object detection on COCO dataset. This study explored a new technique for analyzing heart rhythm data (ECG) to identify both COVID-19 and other heart problems. The method achieved impressive accuracy in distinguishing healthy hearts from those with COVID-19 or other conditions. This has the potential to lead to improved methods for diagnosing these issues. K. Divya et. al [11] introduce a novel convolutional network architecture with two layers sharing the same feature-map size, DenseNets naturally scale to hundreds of layers without optimization issues. The connectivity rule in DenseNets integrates identity mappings, deep supervision, and diversified depth, enabling feature reuse and learning more compact models. It focuses Inception-v3 achieves state-of-the-art performance with a modest 2.5x increase in computational cost compared to previous versions. Our model outperforms denser networks, reducing top-5 (top-1) error by 25% (14%) while being six times cheaper computationally. An ensemble of four Inception-v3 models achieves a 25% reduction in top-5 error, outperforming ILSVRC 2014-winning GoogLeNet ensemble. Naz et. al [12] developed a new way to organize and analyze heart rhythm data (ECG) for use in computer programs. This method involves rearranging specific parts of the data into a format that can be easily processed by image recognition software. This approach is particularly helpful for situations where there's limited data available for certain heart conditions. The researchers found that their method was very accurate (nearly 98%) at identifying different heart rhythm patterns. This suggests that the new method could be a valuable tool for improving the diagnosis of heart problems. Usha et. al [13]

says Early myocardial infarction (MI) detection is crucial, advocating for regular health checkups. With datasets containing and excluding noise, the system achieved impressive average accuracy, sensitivity, and specificity of

94.19%, respectively. Kanksha et. al [14] introduces Traditional and modern techniques, including CNN and deep learning, are employed for ventricular tachyarrhythmia (VTA)

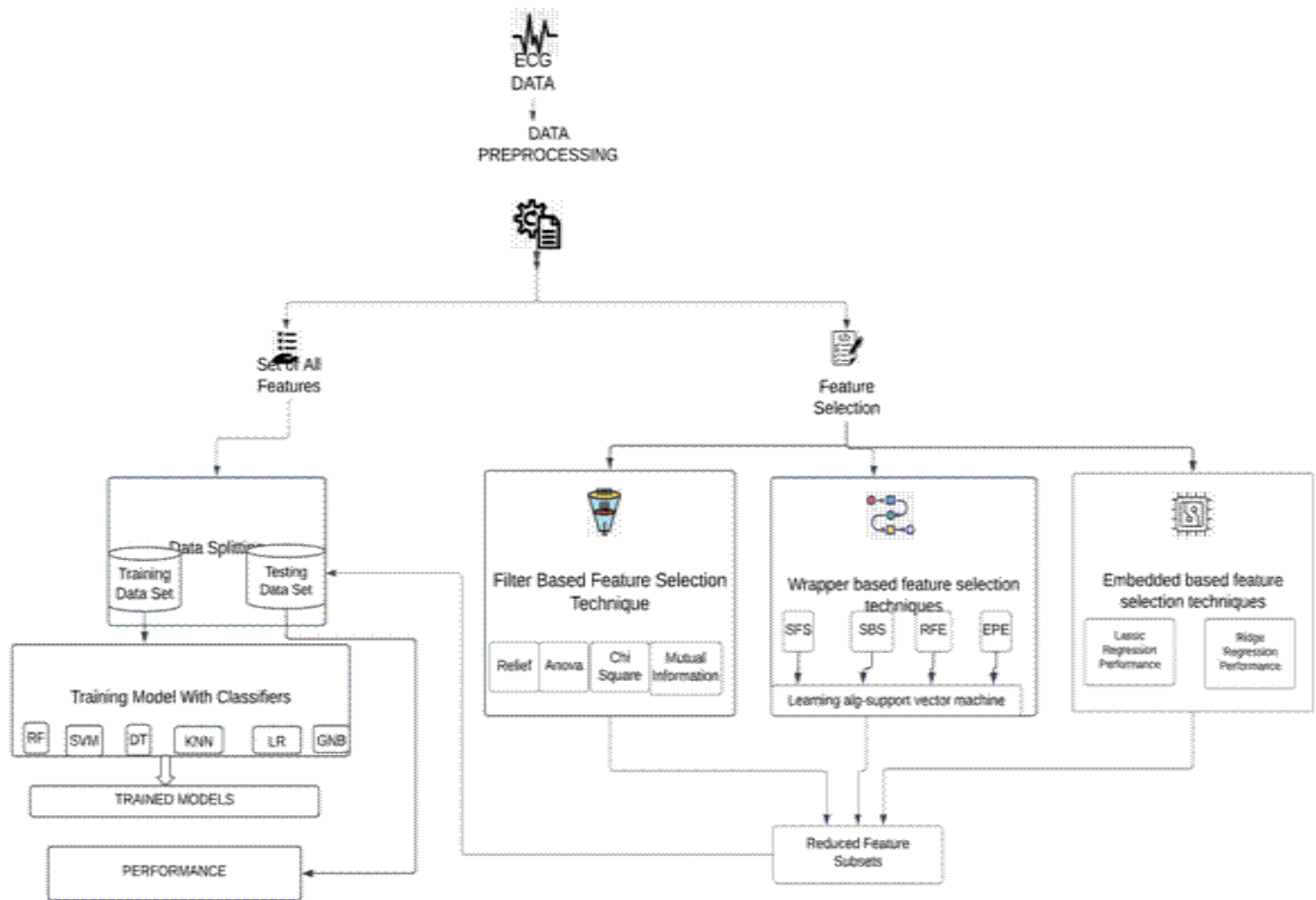


Fig. 1. Prediction Work Flow

93.53%, 93.71%, and 92.83%, and 95.22%, 95.49%, and

automatically identifies a specific heart rhythm pattern (VTA) by converting the ECG data into black and white images. This approach is faster than existing methods and achieved a high success rate (nearly 98%) in identifying VTA in a large database of ECG recordings. This paves the way for improved methods for analyzing heart rhythm data.

III. METHODOLOGY

A. Feature Selection

1) Filter- based Feature Selection Techniques

Data analysis(Fig.1) can get cluttered with too much information. Filter-based feature selection techniques help sort through this data by identifying the most useful pieces. Imagine sorting items by color - these techniques pick the features that best distinguish red from blue items.

2) Wrapper based feature selection techniques

Wrapper-based feature selection methods take a different approach than filter methods. Instead of analyzing features independently, they treat feature selection as part of the model building process itself. Imagine building a puzzle - wrapper methods test different combinations of pieces (features) to see which combination creates the best overall picture (model). There are several ways to do this. Sequential Forward Selection (SFS) starts with no pieces and adds the most helpful one at each step.

3) Embedded based feature selection techniques

Embedded feature selection techniques(Fig3.1) integrate feature selection into the model training process, automatically selecting relevant features during model development. These methods, such as LASSO (Least Absolute Shrinkage and Selection Operator) or tree-based algorithms like Random Forests, incorporate feature importance measures as part of their optimization. Lasso regression is an embedded feature selection technique that combines regularization and linear

regression. This sparsity-inducing property effectively selects relevant features during model training, making Lasso a powerful method for feature selection.

B. MACHINE LEARNING ALGORITHM

1) DECISION TREES

Decision trees offer a significant advantage in their ability to analyze ECG images efficiently. By asking a series of targeted questions about specific features within the image, they can rapidly classify the ECG as indicative of CVD or not.

2) SVM

SVMs are supervised learning algorithms that excel at finding the optimal hyperplane (a separation boundary) in high-dimensional data spaces. They achieve this by identifying a small number of critical data points, called support vectors, that define the margin between different classes. This margin represents the confidence level of the classification.

3) RANDOM FOREST

A new method for analyzing heart rhythm data (ECG) uses a combination of simplified models to improve the accuracy of cardiovascular disease (CVD) diagnosis. This approach has the potential to streamline the process and make CVD diagnosis more accurate.

4) CL

CL algorithms typically deal with data streams where information is presented sequentially, and the model needs to learn and adapt continuously. ECG image analysis usually involves a static dataset of ECG images used to train the model for classification.

5) NB

Naive Bayes (GNB) could help create new features to enhance the overall cardiovascular disease diagnosis system built with deep learning techniques.

While exploring the potential of machine learning algorithms like Support Vector Machines (SVMs) and Decision Trees for ECG image analysis is valuable, their limitations with complex image data require careful consideration. Here investigate feature engineering techniques to extract meaningful information from ECG images suitable for these algorithms. This exploration aims to understand the potential of traditional machine learning for cardiovascular disease diagnosis alongside future investigations into deep learning approaches.

ECG images pose a challenge due to their complex, spatial patterns. Our approach will focus on meticulous feature engineering. By extracting informative features from the images, like statistical measures or segmented regions of interest, we can transform the data into a format suitable for these algorithms. Additionally, exploring ensemble methods that combine multiple models can potentially improve the generalizability and reliability of the predictions

IV. RESULTS & DISCUSSION

Table I investigates the applicability of various data mining and machine learning models on different datasets. Datasets are identified by source (Cleveland Database, UCI repository,

publicly available heart disease dataset, Kaggle) and the number of input attributes they contain.

TABLE I. COMPARISON OF GIVEN DATASETS

S.No	Data-Set	No of Input Attributes	Data Mining/ Machine Learning				
			DT	NB	CL	RB	SVM
1	Cleveland Database	13	√	√			
2		14		√	√	√	
3		15	√	√			
4	UCI repository	13	√		√		√
5	Publicly available heart	13	√	√			
6	Disease Dataset	6	√	√	√		
7	Kaggle						√

*DT-Decision Tree,NB-Naive Bayes, CL-Clustering,RB-Rule Based,SVM-Support Vector Machines

TABLE II. COMPARISON OF EVALUATION METRICES WITH DEPLOYMENT MODELS.

MODELS	TOOLS USED		
	WEKA	TANGARA	MATLAB
DT	√	√	√
NB	√	√	
CL	√	√	
RF	√		
SVM		√	√

The table II explores the software tools suitable for implementing various data mining and machine learning models. The models (Decision Tree (DT), Naive Bayes (NB), Clustering (CL), Rule-Based (RB), and Support Vector Machine (SVM)) are listed on the left, while the top row showcases the software tools (WEKA, TANGO, MATLAB)

TABLE III. PROS AND CONS OF WEKA

S.No	Evaluation	Algorithms				
		DT	NB	CL	RB	SVM
1	Sensitivity	√	√		√	√
2	Specificity	√			√	√
3	Accuracy	√	√	√	√	√
4	Precision	√	√		√	

5	Confusion Matrix	✓	✓			
6	Efficiency	✓				
7	Lift Chart	✓	✓			

TABLE IV. ALGORITHM PERFORMANCE IN ECG IMAGE ANALYSIS FOR CVD DIAGNOSIS

S.No	Machine Learning Algorithm	Accuracy Level	Performance Time
1	Decision Tree	59.36%	550ms
2	Naïve Bayes	56.33%	620ms
3	Clustering	48.90%	1000ms
4	Ruled Based	57.48%	582ms
5	SVM	55%	733ms

Table III provides a helpful overview of how to evaluate different data mining and machine learning models. The top row lists various evaluation metrics (Sensitivity, Specificity, Accuracy, etc.), Table IV summarizes the performance of several machine learning algorithms, likely for disease diagnosis. Decision Tree stands out with the highest accuracy (59.36%) and fastest processing time (550ms) compared to Naive Bayes, Clustering, Rule-Based, and Support Vector Machines

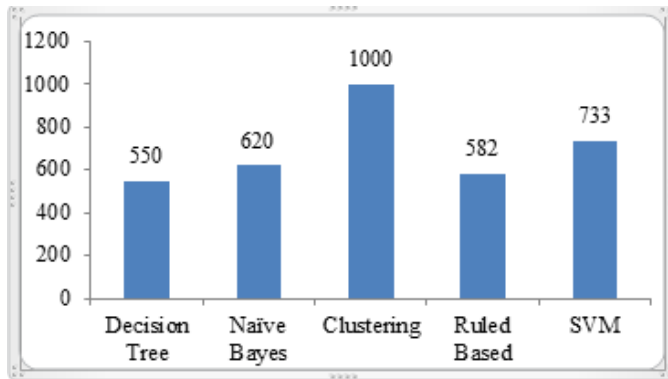


Fig. 2. CVD Diagnosis Performance Timing

Fig.2 visualizes the training efficiency of various machine learning algorithms, highlighting how long each takes to complete the meningitis prediction task using CSF analysis data.

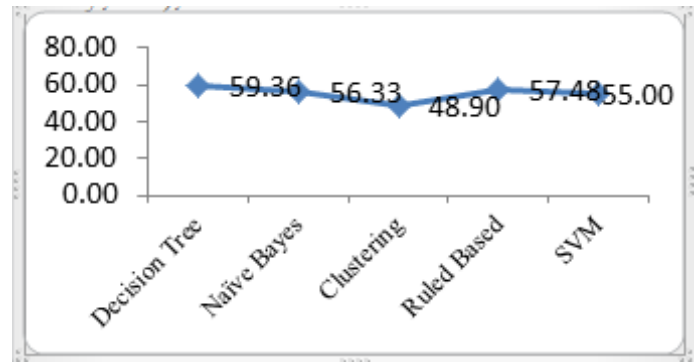


Fig. 3. CVD Diagnosis Accuracy Level

This Fig 3. unveils the effectiveness of various machine learning algorithms for meningitis prediction using CSF analysis data. It highlights the percentage of predictions each algorithm gets correct.

IV. CONCLUSION

This research exploring a variety of statistical methods, including decision trees, probability calculations, and nearest neighbor comparisons, to analyze electrocardiogram (ECG) images and improve cardiovascular disease (CVD) diagnosis. However, their limitations include feature engineering and handling complex data patterns. Deep learning, particularly Convolutional Neural Networks, hold immense potential for the future. Their ability to automatically learn features, model intricate relationships, and work with high-dimensional data paves the way for more accurate CVD detection. Combining the strengths of deep learning with traditional approaches can unlock significant advancements in automated CVD diagnosis using ECG images, leading to earlier interventions, better patient outcomes, and a more efficient healthcare system. Machine learning offers promise for ECG image analysis in CVD diagnosis. While initial exploration might involve Decision Trees, future enhancements lie in deep learning architectures specifically designed for ECG images. Techniques like data augmentation, transfer learning, and multimodal learning can further improve accuracy. These advancements hold the potential to revolutionize CVD diagnosis using ECG images analysis. While exploring Support Vector Machines (SVMs), Naive Bayes (NB), and Decision Trees offers valuable insights, their effectiveness for ECG image analysis in diagnosing cardiovascular diseases might be limited. These algorithms excel at analyzing structured data, but ECG images present complex spatial patterns. This investigation will focus on feature engineering techniques. By meticulously extracting informative features from the images, such as statistical properties or segmented regions, this can transform the data into a format suitable for these algorithms.

REFERENCES

- [1] J. K. Subhadra, B. Vikas, Neural network based intelligent system for predicting heart disease, *Int. J. Innov. Technol. Explor. Eng. (IJITEE)* (ISSN: 2278-3075) 8 (5) (2019)
- [2] K. Dissanayake and M. G. Md Johar, "Comparative study on heart disease prediction using feature selection techniques on classification algorithms," *Appl. Comput. Intell. Soft Comput.*, vol. 2021, 2021, Art. no. 5581806. [Online]. Available: <https://doi.org/10.1155/2021/5581806>.
- [3] A. H. Gonsalves, F. Thabtah, R. M. A. Mohammad, and G. Singh, "Prediction of coronary heart disease using machine learning: 2019, pp. 51–56. [Online]. Available: <https://doi.org/10.1145/3342999.3343015>
- [4] Soumonos Mukherjee, Anshul Sharma, Intelligent heart disease prediction using neural network, *Int. J. Recent Technol. Eng. (IJRTE)* (ISSN: 2277-3878) 7 (5) (2019).
- [5] R. Bharti, A. Khamparia, M. Shabaz, G. Dhiman, S. Pande, and P. Singh, "Prediction of heart disease using a combination of machine learning and deep learning," *Comput. Intell. Neurosci.*, vol. 2021, 2021, Art. no. 8387680. [Online]. Available: <https://doi.org/10.1155/2021/8387680>
- [6] T. Rahman et al., "COV-ECGNET: COVID-19 detection using ECG trace images with deep convolutional neural network," 2021, arXiv:2106.00436.
- [7] T. Kujani, B. R. S. T. P. Arivubakan, V. Usha and T. R, "Quantum Resnet18 for Classifying and Predicting the Maize Leaf Diseases," 2024 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI), Chennai, India, 2024, pp. 1-6, doi: 10.1109/RAEEUCCI61380.2024.10547895.
- [8] T. Kujani and V. D. Kumar, "Ensembled Transfer Learning Models for Real Time Face Recognition," 2022 1st IEEE International Conference on Industrial Electronics: Developments & Applications (ICIDEA), 2022, pp. 35-40, doi: 10.1109/ICIDEA53933.2022.9970195.
- [9] Hindawi Computational Intelligence and Neuroscience Volume 2021, Article ID 8387680, 11 pages <https://doi.org/10.1155/2021/8387680R>.
- [10] Avanzato and F. Beritelli, "Automatic ECG diagnosis using convolutional neural network," *Electronics*, vol. 9, no. 6, 2020, Art. no. 951. [Online]. Available: <https://doi.org/10.3390/electronics9060951>
- [11] K. Divya, A. Sirohi, S. Pande, and R. Malik, "An IoT-assisted heart disease diagnostic system using machine learning techniques," in A. E. Hassanien, A. Khamparia, D. Gupta, K. Shankar, and A. Slowik, Eds., vol. 311, pp. 145–161, Springer, Cham, Switzerland, 2021
- [12] M. Naz, J. H. Shah, M. A. Khan, M. Sharif, M. Raza, and R. Damaševičius, "From ECG signals to images" *PeerJ Comput. Sci.*, vol. 7, 2021, Art. no. e386, doi: 10.7717/peerj-cs.386.
- [13] Usha, V. and Rajalakshmi, N. R., 2023, September. Insights into diabetes Prediction: A Multi-Algorithm Machine Learning Analysis. In 2023 4th International Conference on Smart Electronics and Communication (ICOSEC) (pp. 1207-1212). IEEE.
- [14] Kanksha, B. Aman, P. Sagar, M. Rahul, and K. Aditya, "An intelligent unsupervised technique for fraud detection in health care systems," *Intelligent Decision Technologies*, vol. 15, no. 1, pp. 127–139, 2021
- [15] A. K. Garate-Escamila, A. Hajjam El Hassani, and E. Andres, "Classification models for heart disease prediction using feature selection and PCA," *Informatics in Medicine Unlocked*, vol. 19, Article ID 100330, 2020..